

Majorana Modes in the Machine

Circuit-Level Gate Noise on the VQE: Channels, Verification, and the Depth Optimum

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From a Validated VQE to a Noisy Device

What we already have

- A validated parity-constrained VQE: optimal $\theta^*(\mu)$ across the phase diagram, benchmarked against exact diagonalization.
- A noiseless depth diagnostic: subspace infidelity *falls* as the ansatz repetitions r grow.
- Conclusion so far: a deeper ansatz prepares a better state.

Same Hamiltonian and edge string $O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$ as before; the new ingredient is **noise that enters through the gates**.

The question now

- Put *real gate noise* into that preparation circuit.
- Each repetition adds an entangling layer; every CNOT carries an error channel.
- As the circuit deepens, does the edge-string diagnostic survive?

Gate Errors as Quantum Channels

A noisy gate is the ideal unitary followed by a CPTP error channel:

$$\tilde{\mathcal{U}}_g = \mathcal{E}_g \circ \mathcal{U}_g, \quad \mathcal{E}_g(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I. \quad (1)$$

A full circuit composes these gate-by-gate:

$$\rho_{\text{out}} = (\mathcal{E}_{g_M} \circ \mathcal{U}_{g_M}) \circ \cdots \circ (\mathcal{E}_{g_1} \circ \mathcal{U}_{g_1})(\rho_{\text{in}}). \quad (2)$$

Key consequence

Noise enters *through the gates*: the number of error channels grows with the gate count, hence with the circuit depth. A density-matrix simulation with a gate-level `NoiseModel` captures this exactly.

The Channels, and Which Ones Respect Parity

Integrating the Lindblad dynamics over a gate duration τ turns physical decoherence into CPTP channels. What matters for the Majorana diagnostic is how each acts on parity $\hat{P} = \prod_j Z_j$.

Channel	Physical origin	Effect on the diagnostic
Depolarizing	isotropic, symmetric noise	shrinks all correlations / string contrast
Amplitude damping (T_1)	relaxation $ 1\rangle \rightarrow 0\rangle$	parity-violating : leaks into the odd sector
Phase damping (T_2)	loss of phase coherence	parity-preserving : kills endpoint coherence
Thermal relaxation	T_1 and T_2 combined	the realistic device channel
Readout	bit-flip at measurement	biases the ± 1 parity-product estimator

Two effects are qualitatively distinct: T_2 erodes the *contrast* of the non-local string, while T_1 *poisons the parity sector* the diagnostic lives in.

Noise Integrated Through the Gates, by Depth

The linear EfficientSU2 ansatz at depth r has

$$\# \text{ CNOTs} = r(L - 1),$$

and each one applies a depolarizing channel. The *measured* signal is a product of two opposing factors:

$$|\langle O_{\text{edge}} \rangle|_{\text{meas}}(r) \approx \underbrace{|\langle O_{\text{edge}} \rangle|_{\text{ideal}}(r)}_{\text{expressibility}} \times \underbrace{(1 - p_{\text{cx}})^{r(L-1)}}_{\text{accumulated gate noise} \downarrow}. \quad (3)$$

- Expressibility has a **threshold**: below a minimum depth the ansatz cannot represent the state ($|O|_{\text{ideal}} \approx 0$); above it, $|O|_{\text{ideal}}$ saturates near 1.
- Gate noise then *falls* exponentially in depth.

The depth trade-off

The optimum is the **shallowest adequately-expressive depth**: just deep enough to prepare the state, no deeper, because every extra layer only adds CNOT noise.

Integrating Gate Noise in Qiskit

```
from qiskit_aer import AerSimulator
from qiskit_aer.noise import NoiseModel, depolarizing_error

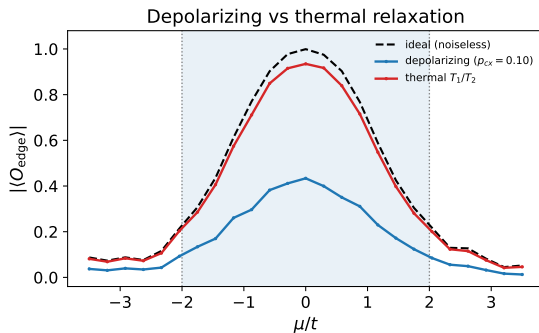
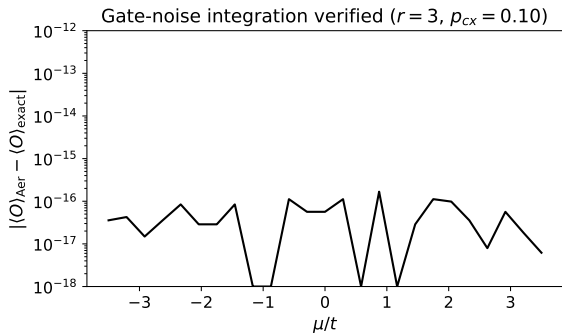
nm = NoiseModel()
nm.add_all_qubit_quantum_error(depolarizing_error(p_cx, 2), ['cx'])

sim = AerSimulator(method='density_matrix', noise_model=nm)
qc = ansatz.assign_parameters(theta)    # the validated VQE state prep
qc.save_density_matrix()
rho = sim.run(transpile(qc, sim)).result().data()['density_matrix']
edge = rho.expectation_value(0_edge).real
```

- `add_all_qubit_quantum_error` fires the channel after every `cx` in the transpiled circuit.
- We transpile to a fixed basis `['rz', 'sx', 'cx']` so the `cx` the error attaches to is exactly the `cx` the circuit runs.
- Reading $\text{Tr}[O_{\text{edge}}\rho]$ from the density matrix avoids shot noise while we study the gate layer.

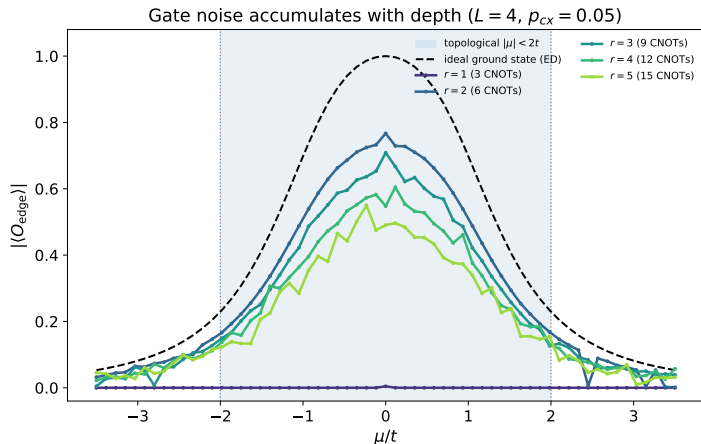
Precise Verification + a Physical T_1/T_2 Model

Verified circuit-level gate noise and a physical T_1/T_2 model



Left: the Aer NoiseModel edge string vs. an *independent* exact gate-by-gate density-matrix evolution (same circuit, depolarizing SuperOp after each cx) agree to $\sim 10^{-15}$ across all μ – the gate noise is correctly integrated through the whole circuit. **Right:** swapping depolarizing for `thermal_relaxation_error( $T_1, T_2, \tau$ )` gives the realistic channel; T_1 additionally poisons the parity sector, T_2 kills the endpoint coherence the edge string needs.

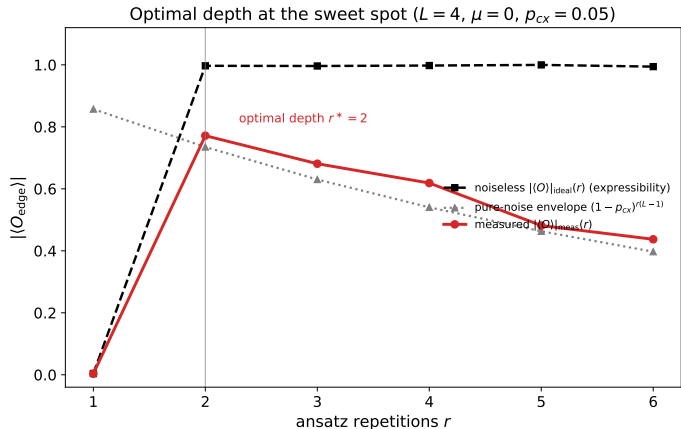
Gate Noise vs. Depth Across the Phase Diagram



Same per-cx strength at every depth; only the CNOT count $r(L - 1)$ changes.

- $r = 1$ is *under-expressive*: the diagnostic is dead everywhere (flat at zero).
- $r = 2$ first reaches the state, then deeper circuits are pushed *down* by accumulated CNOT noise.
- The phase contrast is sharpest at the $\mu = 0$ sweet spot and shallow depth.

The Depth Optimum, Made Explicit



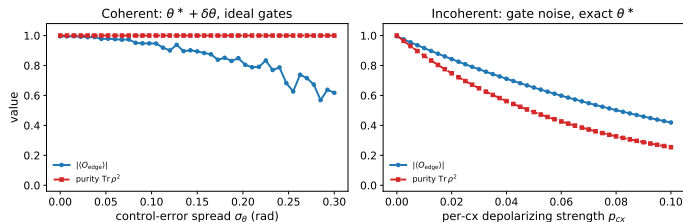
At the $\mu = 0$ sweet spot, the two factors separate:

- **Expressibility** (dashed): a step – zero at $r = 1$, saturated for $r \geq 2$.
- **Pure-noise envelope** (dotted): smooth $(1 - p_{cx})^{r(L-1)}$ decay.
- **Measured** (solid): $\approx$ their product, peaks at $r^* = 2$.

The optimum is not a smooth balance – it is the *first* depth that can express the state.

Two Families of Error: Coherent vs. Incoherent

Coherent control error rotates a pure state; incoherent noise mixes it ($L = 4$, $r = 3$, $\mu = 0$)



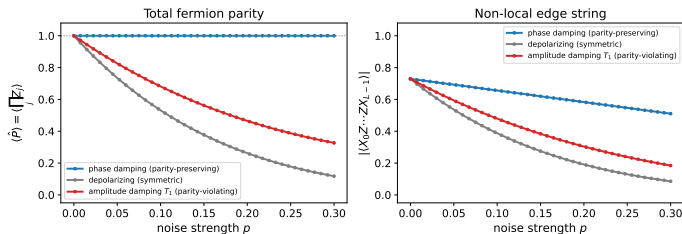
Same prepared state, two error types:

- **Coherent** ($\theta^* + \delta\theta$):
miscalibrated angles, ideal gates.
The state stays *pure* ($\text{Tr} \rho^2 = 1$) – only rotated off target.
- **Incoherent** (per-cx depolarizing): genuine decoherence – signal *and* purity fall.

A coherent error can be re-absorbed by recalibration or the VQE; incoherent loss cannot.

Professor's Q1: Is the Parity Protected by Topology?

Parity (symmetry) survives dephasing; the edge string (topology) does not ($L = 6$, $\mu = 1t$)



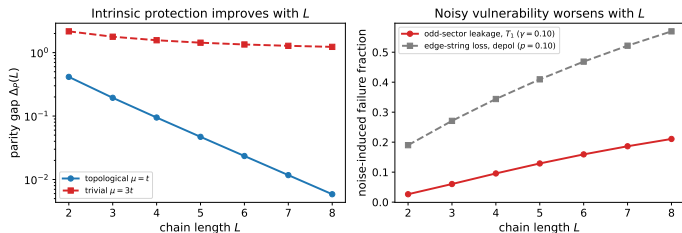
Topological state ($L = 6$, $\mu = t$).

- **Left:** $\langle \hat{P} \rangle$ stays *exactly* 1 under phase damping – but that is the $\mathbb{Z}_2$ symmetry (true in every phase), **not** topology. It collapses under depolarizing, leaks under T_1 .
- **Right:** the topological edge string decays under *every* channel – even the parity-preserving one.

Answer: parity is symmetry-protected; the topological signal is not noise-immune at fixed L .

Professor's Q2: How Does the Chain Length Matter?

Chain length: opposite effects under noise



Length acts in *opposite* directions:

- **Left:** the intrinsic parity gap is protected exponentially, $\Delta_P(L) \sim e^{-L/\xi}$ – longer chains are intrinsically *better*.
- **Right:** at fixed noise, T_1 odd-sector leakage and depolarizing edge-string loss both *grow* with L – longer chains are *noisier* to operate.

Answer: the two trends oppose, defining an optimal working length – the chain-length analogue of the optimal depth.

Summary and What This Enables Next

- Gate noise is integrated at the *circuit* level via a Qiskit NoiseModel, verified to machine precision against an exact gate-by-gate reference.
- The diagnostic is bounded by an expressibility threshold, then eroded by accumulated CNOT noise: the optimum is the **shallowest adequately-expressive depth**.
- Parity is protected by *symmetry*, not topology; the topological signal is not noise-immune at fixed L , and longer chains help the gap but hurt the device.

Next

- **Noisy VQE optimization**: a NoiseModel-backed estimator, so every cost evaluation is corrupted.
- **Backend-calibrated noise**: NoiseModel.from_backend(...) for real T_1/T_2 and gate-error data.

Code: src/block4.py (plots 2–7).

Notes: week9_note.tex, parity_topology_protection.tex.